An Algorithm for Detecting Coherence in Cyclone Global Navigation Satellite System Mission Level-1 Delay-Doppler Maps

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***Abstract*— An algorithm for detecting coherence in Cyclone Global Navigation Satellite System (CYGNSS) mission delay-Doppler maps (DDMs) is presented. Because CYGNSS DDMs report only the observed power without phase information, the algorithm uses estimates of power "spread" within the DDM to flag coherency. Since the estimate used is a ratio of the powers in differing portions of the DDM, it is less sensitive to absolute power calibration and to the GPS C/A code type observed, and is applied to CYGNSS Level-1 uncalibrated DDMs. The basic detector formulation is described along with modifications to improve performance in lower signal-to-noise ratio (SNR) situations. The required detection thresholds are determined using matchups with CYGNSS "Raw I/F" mode measurements for which the DDM phase can be computed and used to identify coherence more precisely. Application of the final detector over a large CYGNSS data set suggests that approximately 8.9% of all inland returns are coherent. Inland regions persistently identified as coherent were found largely to be associated with the presence of water bodies. A smaller set of desert locations apparently having very low surface roughness were also found to be associated with persistent coherence. The detector was also applied to a set of ocean measurements, with the results showing that persistent coherence is limited to areas with sheltered waters. Ocean tests avoiding such regions indicate that the detector's false-alarm rate is approximately 0.0012% for the detection threshold used.**

***Index Terms*— Bistatic radar systems, coherency, Cyclone Global Navigation Satellite System (CYGNSS), Global Navigation Satellite Systems Reflectometry (GNSS-R), rough surface scattering.**

I. Introduction

GLOBAL Navigation Satellite System Reflectometry (GNSS-R) is an emerging technology for Earth remote sensing [1]–[6]. The Cyclone Global Navigation Satellite System (CYGNSS) mission has provided a data set spanning more than two years of measurements over both ocean and land surfaces to demonstrate GNSS-R sensing methods. While CYGNSS's primary mission objective is the measurement of ocean wind speeds [7]–[11], CYGNSS measurements are now being applied for land applications including soil moisture [15]–[20] and biomass retrievals [21]. The high sensitivity of CYGNSS inland measurements to the presence of water bodies has also motivated studies of flood detection, the measurement of wetland extent, and inland water body mapping [22]–[26]. Over land, the bistatic specular forward scatter CYGNSS observes may be dominated by coherent reflection or incoherent scattering, with the influence of these contributions determined by a complex combination of terrain properties [28] such as topography, small-scale surface roughness, and the presence of inland water bodies. Because coherent reflection and incoherent scattering are differing physical effects that have distinct relationships to scene and sensor properties, it is important to distinguish these effects in measured data so that retrieval methods can be developed based on the applicable physical process. The development of methods to identify CYGNSS measurements that are dominated by coherent or incoherent effects is, therefore, important for supporting science investigations over Earth's land surface.

This article describes an algorithm for detecting coherence in CYGNSS measurements that can be applied for both land and ocean measurements. Because coherent and incoherent returns show differing "spreads" in CYGNSS delay-Doppler map (DDM) measurements, the algorithm detects coherence using the extent of power spread across a DDM. Detection thresholds are developed through comparisons with CYGNSS "Raw I/F"-mode measurements, which are capable of measuring signal phase and, therefore, provide a more unambiguous detector for coherence. The performance of the detector and selected thresholds is then analyzed over a set of test sites and also over a larger set of CYGNSS measurements.

Section II provides an overview of CYGNSS Level-1 data products, and Section III describes the detection algorithm developed. It then compares detection results against Raw I/F-mode results as well as detections using the global CYGNSS data set. The dependence of the results obtained on the prevalence of water bodies, varying levels of surface roughness, and signal-to-noise ratios (SNRs), along with results from applying the method to ocean observations, are then

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explored in Section IV. Section V provides a summary of this study's findings and provides recommendations for future work concerned with the detection of coherence at a global scale using spaceborne GNSS-R systems.

II. CYGNSS Data Description

The CYGNSS mission provides three levels of data products, Levels 1–3, as publicly available netCDF files. The proposed detection methodology applies to Level-1 (L1) data, from which the latter two data products are derived.

A. Standard Level-1 Data

The fundamental measurement that CYGNSS provides is the DDM that results from the mapping of received power from the observed surface to delay-Doppler space. Delay diversity arises due to the varied paths received signals taken from the scattering surface point to the CYGNSS observatories, while Doppler diversity is due to the relative motions of the GPS transmitter, CYGNSS receiver, and the Earth's surface. L1 DDMs are cropped from a larger "full" DDM formed on board the CYGNSS delay-Doppler Mapping Instrument (DDMI) and have a 17-bin extent in delay at $\approx$0.25-μs reporting interval and an 11-bin extent in Doppler at $\approx$500-Hz reporting interval [11]. The first three to four delay rows typically represent thermal noise, so that an L1 DDM has a 3.5 chip delay extent, on average, allowing each measurement to span an approximate surface area ranging from 70×70 km^2 to 100×100 km^2 depending on incidence angle. Originally, each L1 DDM was the result of a 1-ms coherent integration followed by a 1000 look incoherent integration that reduces the effects of speckle noise. This together with the fact that each of CYGNSS's eight observatories is capable of tracking four reflections simultaneously allowed the constellation to provide 32 DDM measurements per second. Beginning July 2019, the CYGNSS incoherent integration time was reduced to 0.5 s so that eight DDMs are reported per satellite per second. This change is particularly beneficial to land studies [15]–[22] as it reduces the along-track spatial "smearing" of DDMs during the integration period.

Downlinked L1 DDMs are in uncalibrated instrument "raw counts" that later undergo two levels of calibration [29], [30]. The calibration converts Level-0 raw counts into power in watts and later to bistatic radar cross section (BRCS) by correcting for the effects of thermal noise, instrument gain, antenna patterns, and propagation losses. For a more comprehensive overview, see [31]. Due to varied uncertainties associated with the calibration process, this study utilizes DDMs in raw counts. This is particularly beneficial, as it bypasses the need for absolute power calibration, whose improvement remains an ongoing area of active development [32]. The results to be shown are also not expected to be affected by pseudo-random-noise (PRN) code specific variations in the width of the GPS L1 C/A code-delay-spreading function recently highlighted in [33].

B. Improved Land CYGNSS Test Data Set

In support of studies exploring the utility of the CYGNSS constellation for land remote sensing, future releases of the data will include several improvements. In addition to improvements in CYGNSS antenna pattern estimates and the GPS effective isotropic radiated power (EIRP), an improved land specular point location solver will be incorporated. This is of particular importance as the existing specular point location algorithm uses only the DTU10 digital elevation model and, therefore, does not correct for topographic variations. For specular points at high elevation, this may result in appreciable errors in geolocation. To examine these improvements, a special test set called a "sandbox run" was created for a 12-day-period spanning day of year (DOY) 240–251 in 2019. The test run provides added ancillary information not typically incorporated as part of L1 data, including estimates of static water body fraction within a measurement's footprint as well as terrain elevation and slopes obtained from Shuttle Radar Topography Mission (SRTM) digital elevation map. The sandbox run includes DDMs for $\approx$16.5 million specular points over land and $\approx$45 million specular points over the ocean across all eight CYGNSS receivers. This data set is used in what follows for testing the proposed coherence-detection algorithm.

III. Coherence-Detection Algorithm

Several previous studies have attempted to develop techniques through which coherence may be detected as a means of distinguishing sea ice from sea surface measurements. Many of the methods developed rely on mean values within the DDM average (DDMA) region within varying delay-Doppler extents [34], [35], or on SNRs [36], [37], or on the DDM trailing edge slope [34], [38], correlations [34], or a combination of these approaches. This work proposes a method related to the extent of power spread across the DDM and compares with a subset of previously developed methods.

A. DDM Power-Spread-Based Detection

Coherent returns typically arise due to a mirror-like reflection of the transmitted signal off the Earth's surface. This is due to the smoothness of the surface within the observed footprint, with properties that are explored in [39]. Due to the low level of surface roughness, the coherent received field is reflected from the specular point and the small surface area surrounding it on the order of the first Fresnel zone in size [40] with dimensions varying in accordance with the observing geometry. Nonetheless, it is noted that other studies [41] have highlighted the complexities associated with exact quantification of the spatial extent of this small surface area under a scattering regime dominated by coherent reflections, with similar ambiguities arising due to varying observation geometries [42]. Coherence, therefore, refers to a scattering situation in which returns from large portions of the measurement's footprint arrive at the receiver having identical phase shifts, and therefore add coherently. The DDM is then characterized by a sharp peak at the specular bin and a concentration of power within ± 1 delay bin and ± 2 Doppler bins about the DDM peak. In contrast, rougher surfaces give rise to incoherent returns in which the received power is scattered from points at significantly farther distances from the specular point. In delay-Doppler space, this translates into

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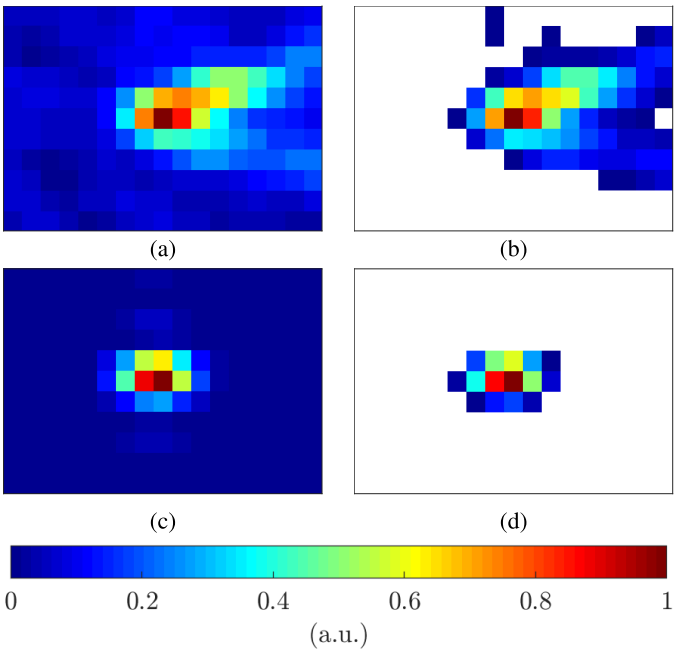


Fig. 1. Illustration of noise pixel-exclusion effects. Regions in white are excluded from the computation of C_{out}. (a) Incoherent CYGNSS L1 DDM—no noise exclusion. (b) Incoherent CYGNSS L1 DDM—with 30% noise exclusion threshold. (c) Coherent CYGNSS L1 DDM—no noise exclusion. (d) Coherent CYGNSS L1 DDM—10% exclusion threshold.

a larger spread in power at delay and Doppler values beyond the specular point.

CYGNSS land measurements show a variety of behaviors, including cases with power spread throughout the DDM as well as mixed DDMs indicative of partial coherence in which the powers from incoherent scattering and coherent reflection are both observable. These mixed cases also have appreciable power spread compared with coherent DDMs. In what follows, incoherent, mixed, and low SNR DDMs will be classified as noncoherent, so that the detected DDMs can be regarded as "purely" or "dominantly" coherent. The proposed DDM power-spread detector (DPSD) compares the power concentrated in the limited delay-Doppler space about the specular bin to the total power outside this region, which is expected to be indicative of the nature of the return due to the correspondence of the physical mechanism of scattering and the extent of power spread. An estimate of power spread is developed through a power ratio (PR)

$$PR = \frac{C_{\text{in}}}{C_{\text{out}}} \tag{1}$$

with

$$C_{\text{in}} = \sum_{i=-1}^{1} \sum_{j=-2}^{2} \text{DDM}(\tau_M + i, f_M + j) \tag{2}$$

$$C_{\text{out}} = \sum_{i=1}^{N_\tau} \sum_{j=1}^{N_f} \text{DDM}(i, j) - C_{\text{in}} \tag{3}$$

where C_{in} is the integrated power in raw counts within the (3×5) delay-Doppler region about the specular bin, C_{out} is the integrated power across the DDM excluding the space

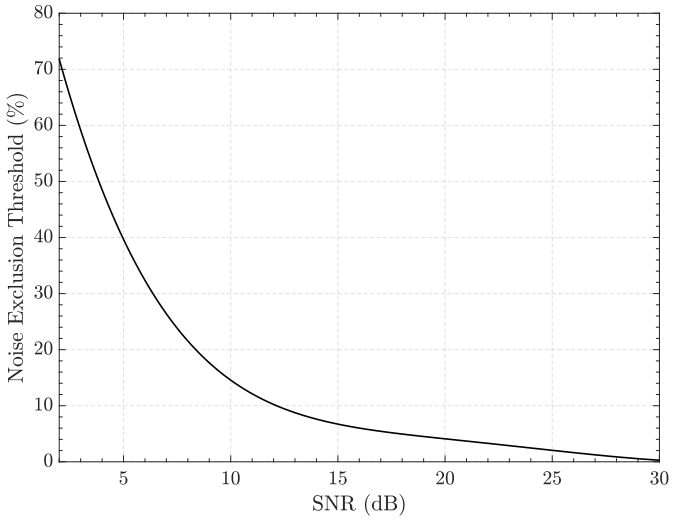


Fig. 2. Noise-exclusion threshold as a function of CYGNSS measurement SNR.

spanned by C_{in}, N_τ is the number of delay bins, N_f is the number of Doppler bins, and τ_M and f_M are the delay-Doppler indices of the specular bin, respectively. Here, τ_M and f_M correspond to the location of the maximum value within a DDM. For coherent returns, the received power will be concentrated within C_{in} with little power in C_{out} so that PR is large. For noncoherent DDMs, the increased spreading of DDM power should result in a lower value for PR. DDMs are flagged as dominantly coherent when $PR \geq \rho_0$ with ρ_0 representing the detection threshold. The idea of power spread across the DDM had been previously explored in [43] but in a manner that focused on absolute power levels within the entire DDM exceeding a preset threshold as opposed to relating power extents within the C_{in} and C_{out} regions.

B. Reducing the Impact of Thermal Noise

The sensitivity of the DPSD method can be improved by refining the computation of C_{out} to reduce the impact of DDM pixels suspected of containing only thermal noise contributions [44]. The mapping of received power from the spatial domain to each pixel within the DDM is dictated by the intersection on Earth's surface of constant delay ellipses with constant Doppler frequency hyperbolas. This process gives rise to the characteristic "horse-shoe" shape of an incoherent DDM. Some portions of the DDM fail to map onto Earth's surface; these DDM pixels contain primarily thermal noise contributions. Because such pixels convey no information on coherence, it is desirable to exclude them from the C_{out} summation. A simple method was, therefore, developed in which pixels having powers less than a predefined percentage of the DDM maximum are excluded from the C_{out} summation.

Example incoherent and coherent DDMs are illustrated in Fig. 1 to clarify this process. Figs. 1(a) and (b) demonstrate an example incoherent DDM in which all DDM pixels outside the specular region or only those at least 0.3 times the DDM maximum are used to compute C_{out}, respectively. The corresponding PR changes from 0.119 to 0.116 in these cases. Figs. 1(c) and (d) similarly show an example coherent

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DDM before and after pixel exclusion, in this case retaining pixels having amplitudes at least 0.1 times the maximum. PR in this case is amplified from 1.437 to 3.543 following the exclusion process. The limited effect on PR of the exclusion for noncoherent DDMs and the significant effect for coherent DDMs motivates the use of this method. Note that an exclusion threshold that is proportional to the DDM maximum value implies that thermal noise may still be retained in low SNR situations. For this reason, the noise-exclusion threshold (the percentage of the DDM maximum that pixel powers must exceed to be included in C_{out}) was specified as a function of the DDM SNR. Tests varying this threshold were used to produce the empirically derived function shown in Fig. 2, which is used in all results to follow.

C. Matched Filter-Based Detection

Results from the DPSD method are compared in what follows with those based on DDM correlation with the CYGNSS Woodward Ambiguity Function (WAF) [16], [34]. The WAF is defined as the DDM response induced by a point scatterer and has the form

$$\chi^2(\tau, f_D) \approx \Lambda^2(\tau) \cdot |S(f_D)|^2 \tag{4}$$

$$|S(f_D)|^2 = \left| \frac{\sin(\pi f_D T_i)}{\pi f_D T_i} \right|^2 \tag{5}$$

$$\Lambda^2(\tau) = \begin{cases} \left(1 - \dfrac{|\tau|}{\tau_c}\right)^2 & |\tau| \le \tau_c \\ 0 & |\tau| > \tau_c \end{cases} \tag{6}$$

where $\Lambda(\tau)$ is the delay-spreading function, $S(f_D)$ is the Doppler-spreading function, T_i is coherent integration time, and τ_c represents the C/A code chip duration of 0.97 μs. The correlation metric is defined here as

$$R = \frac{|\langle \text{DDM}(\tau, f_D), \chi(\tau, f_D) \rangle|^2}{\langle \text{DDM}(\tau, f_D), \text{DDM}(\tau, f_D) \rangle \langle \chi(\tau, f_D), \chi(\tau, f_D) \rangle} \tag{7}$$

in which the bracket notation refers to a pointwise multiplication of DDM "pixels" followed by a summation of all pixel products. The correlation metric R also neglects amplitude information and instead focuses on the overall similarity of the DDM to the WAF. Coherent DDMs are expected to show a high correlation with the WAF that should reduce for incoherent DDMs, and a threshold can be placed on R above which coherency is declared.

D. Setting Detection Thresholds

1) Comparison With Raw I/F-Mode Data: CYGNSS satellites are capable of reporting the observed intermediate frequency raw data stream (in uncalibrated counts) over short periods of time. These "Raw I/F"-mode data can be post-processed into DDMs having arbitrary integration times and delay extents, and can provide DDM phase information that is very sensitive to the presence of coherence. The studies in [45] report an algorithm for detecting coherence in Raw I/F-mode DDMs based on the growth in coherent DDM power as the integration time is extended. Detections obtained from this "gold standard" method were compared with those obtained from the DPSD approach for three reference tracks for which Raw I/F-mode data were available.

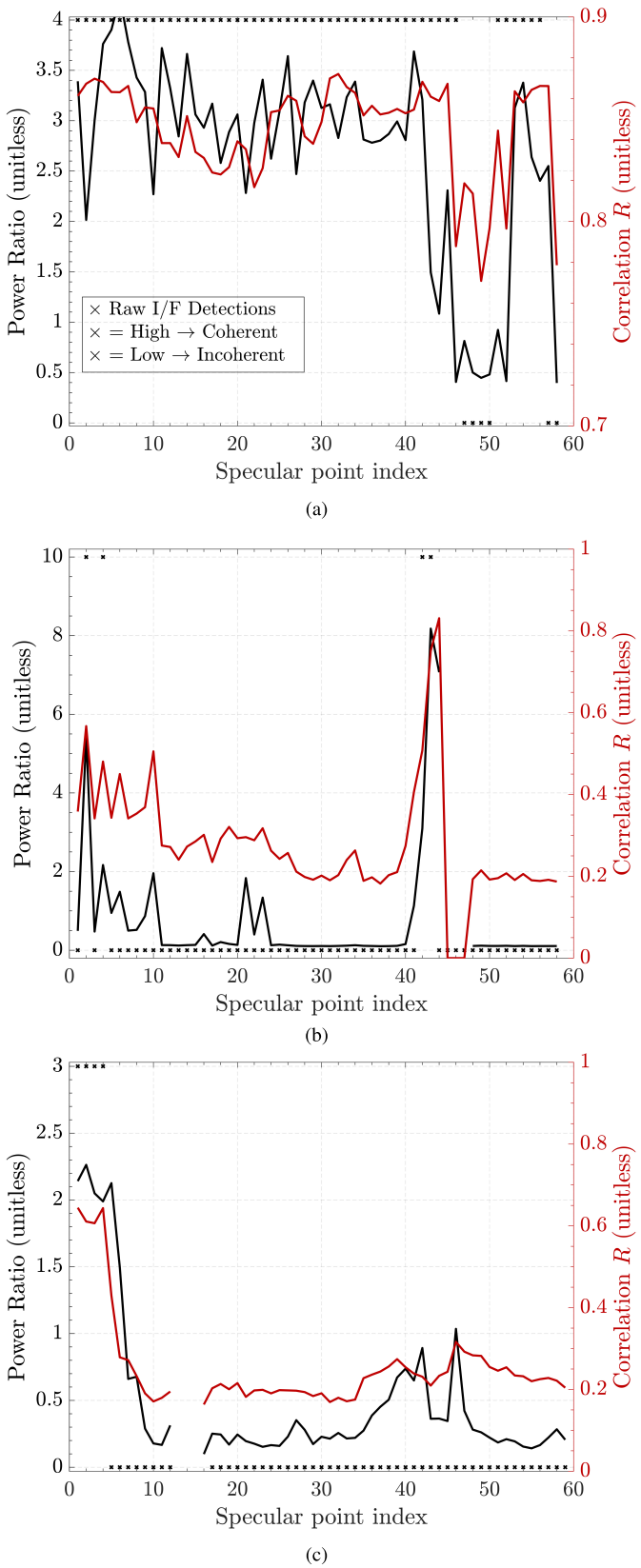


Fig. 3. Comparison of Level-1 DPSD PR and MFD correlation against Raw I/F coherence/noncoherent detection. (a) Track 1, dominantly coherent. (b) Track 2, dominantly noncoherent. (c) Track 3, dominantly noncoherent.

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Fig. 3 illustrates the comparisons. For track 1 (plot a), the gold standard method detected coherence in 53 out of 59 1-s periods (detections are indicated by the $\times$ symbols), with brief intervals of noncoherency occurring in later portions of the 1-min data set. For track 2 (plot b), the gold standard method detects coherence in only 4 out of 59 intervals. For the third track, a brief interval of coherence is detected comprising of four consecutive specular points at the beginning of the track followed by 55 noncoherent DDMs. The PR quantity in plot (a) is found to fall below a value of 2 only for those points flagged as incoherent (with one exception), while the PR quantity is small for the incoherent cases in plot (b). Both the DPSD and matched filter detection (MFD) approaches appear to be able to capture the overall coherence/noncoherence tendencies compared with the Raw I/F detections. However, for track 1, the average PR and R values over coherent intervals are 3.12 and 0.85, respectively, while the noncoherence interval averages are 0.77 (PR) and 0.82 (R), respectively. A similar response is noted for track 3, in which coherent detections were associated with a 2.11 average power ratio, falling below a value of 2 when transitioning to a dominantly noncoherent mode of scattering where the average PR for the noncoherence interval was 0.38. The higher sensitivity and increased separation in PR between dominantly coherent and dominantly incoherent DDMs motivate its selection for use in coherence detection. The results of these tests suggest that ρ_0 should be set to approximately 2 to provide a good match to the three Raw I/F-mode tracks investigated.

2) Land Sandbox Tests: The Raw I/F-mode comparison motivates use of the PR metric to detect coherence. However, the limited Raw I/F-mode data set available does not allow complete determination of the detection threshold ρ_0. DPSD algorithm detections were, therefore, examined, as ρ_0 was varied for the 12-day CYGNSS "sandbox" data set. The resulting percentages of land DDMs detected as coherent are shown in Fig. 4 as a function of ρ_0. The resulting percentages range from $\approx$6.6%–11.4% for ρ_0 ranging from 2.4 to a much more lenient threshold of 1.8. Results from the Raw I/F comparison and manual examinations of numerous DDMs suggest setting the detection threshold ρ_0 as 2.0, with the result that $\approx$8.9% of all inland points, a total of 1.5 million, are declared coherent. It is recognized that there remains some uncertainty in the threshold value used and that a slightly larger or smaller percentage of inland returns would be flagged as coherent if ρ_0 were decreased or increased, respectively. The optimal value for the detection threshold is nonetheless expected to be limited to the range $1.95 \leq \rho_0 \leq 2.05$. Extensive testing showed that the selected 2.0 value appears to provide a good tradeoff between detection sensitivity and false detections.

IV. Results

A. Examples

Example detection results are shown for tracks through Lake Victoria and the Everglades region of Florida in Figs. 5 and 6, respectively. Most specular points within the track over Lake Victoria are classified as noncoherent. The small portion flagged as coherent apparently corresponds to calmer waters close to shore; an example DDM from position 1 is provided in plot 5(c). As the track moves across the lake, typical horseshoe-shaped DDMs indicative of incoherent scatter are observed; this should be expected as the large size of Lake Victoria allows the development of significant wave heights that result in incoherent scattering. An example of the resulting effects is shown in the DDM from position 2 in plot 5(d).

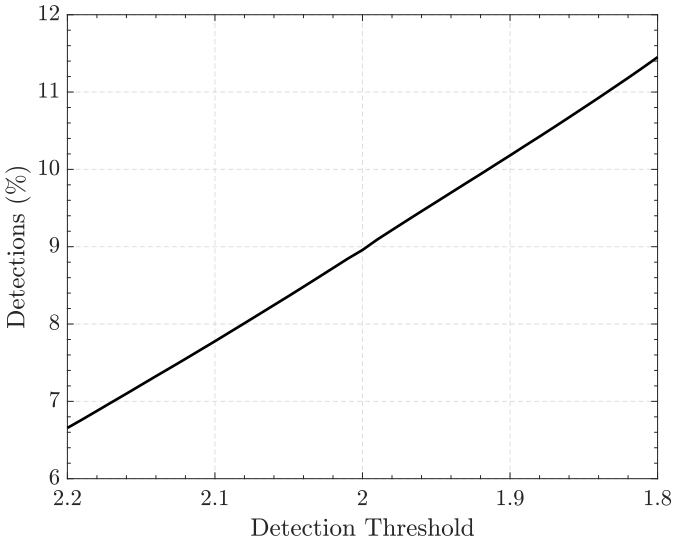


Fig. 4. Percentage of CYGNSS land-surface DDMs detected as coherent for varying ρ_0.

Conversely, the majority of DDMs for the Everglades track were declared coherent, suggesting that scattering surfaces sufficiently flat to allow coherent reflections are widespread in this region. The annual water recurrence map [46] depicted in Fig. 6(b) shows high values in this region, indicating the presence of water bodies that are static or present for most of the year. The coherent returns detected, therefore, are associated with the presence of water bodies. The average PR value for all coherent detections across this track was $\approx$3.48. A limited number of DDMs were classified as noncoherent, all of which correspond to areas with little to no water recurrence. The DDM at position 2 [see Fig. 6(d)] has $PR = 0.99$ and, therefore, is declared noncoherent, but is likely representative of a mixed return with both coherent contributions from water bodies and incoherent scattering from the land surface. Such returns have been found to be associated with high SNR ($>$15 dB) and low PR ($<$2.0). The algorithm developed classifies such returns as noncoherent given the "purely coherent" definition used for the DPSD algorithm.

B. Persistent Coherence Over Land

Specular land surface scattering depends on surface small-scale roughness, large-scale topography, land cover, surface permittivity, and any presence of water bodies. CYGNSS DDMs from some locations will exhibit a higher likelihood of being classified as coherent due to these factors, particularly the presence of water bodies. To investigate these factors, Fig. 7 presents the median PR value over the 12-day sandbox data set following the projection of measured DDMs onto a 36-km Equal-Area Scalable Earth (EASE) grid [47], [48]. The resulting detector output using the median PR value is also shown. The results identify locations that exhibit a higher

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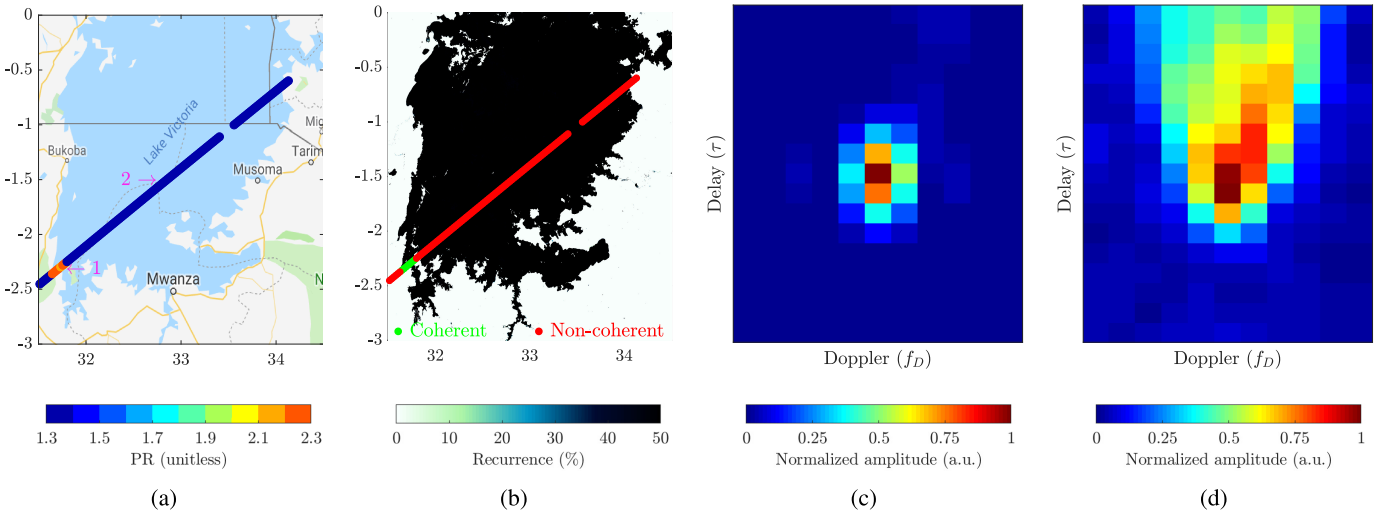


Fig. 5. Case study for CYGNSS L1 coherence detection at Lake Victoria. (a) Power ratios across CYGNSS specular point tracks. (b) Yearly water recurrence percentage. (c) DDM detected as coherent at position 1. DDM SNR is 19.07 dB and its PR is 2.83. (d) DDM detected as noncoherent at position 2. DDM SNR is 5.48 dB and its PR is 0.28.

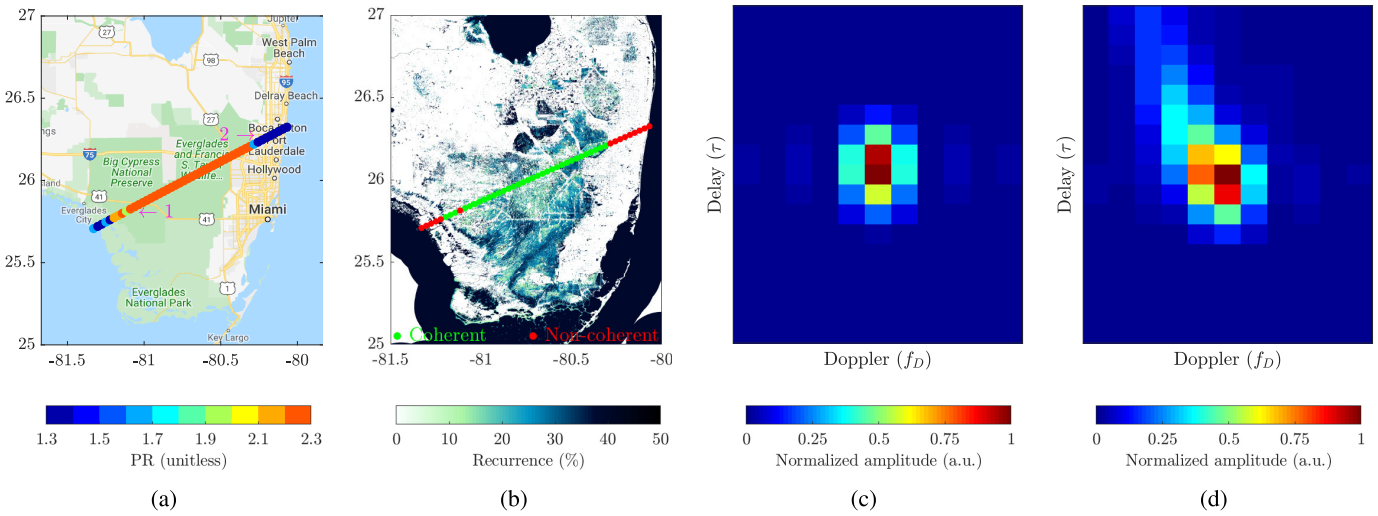


Fig. 6. Case study for CYGNSS L1 coherence detection at the Florida Everglades. (a) Power ratios across CYGNSS specular point tracks. (b) Yearly water recurrence percentage. (c) DDM detected as coherent at position 1. DDM SNR is 23.18 dB and its PR is 2.65. (d) DDM detected as noncoherent at position 2. DDM SNR is 17.73 dB and its PR is 0.99.

likelihood of coherence, including the Florida Everglades, Louisiana Wetlands, and Mississippi River wetlands in the USA, the Orinoco Wetlands in Venezuela, the Iberá Wetlands in Argentina, the vicinity of the Congo River in the DRC, and the Rann of Kutch lake and salt marsh in India/Pakistan. The region bordering the Himalayas extending from Bangladesh through Nepal similarly contains extensive networks of water bodies including the Padma River, Brahmaputra River, and irrigation networks. All these regions have a significant prevalence of water bodies that provide reflecting surfaces sufficiently flat to produce coherent reflections. A further analysis showed that, globally, approximately 81% of DDMs flagged as coherent included a nonzero water fraction within the first Fresnel zone.

Other frequently coherent locations include parts of the Sahelian Zone in Mauritania, the Saharan Zone in Mauritania, the Algerian Desert in Algeria, the Libyan Desert in Libya, and the semiarid savanna in Africa. Water bodies are not widespread in these regions, indicating that the terrain surface must be sufficiently flat to cause coherent reflections. Surface rms heights (including topography within the first Fresnel zone) need to be on the order of approximately 5 cm or less for coherence to be dominant at L-band [39], [49]. An analysis of the SRTM digital elevation model showed that these locations are associated with global rms height minima within a first Fresnel zone-sized region. However, an exact quantification of surface roughness would require measurements made with centimeter (or less) accuracy that, currently, is not available for the locations of interest. The results obtained nevertheless suggest these areas are likely to be sufficiently flat for coherence to be dominant. It is noted that other similar desert regions are not identified as producing highly coherent returns; analyses

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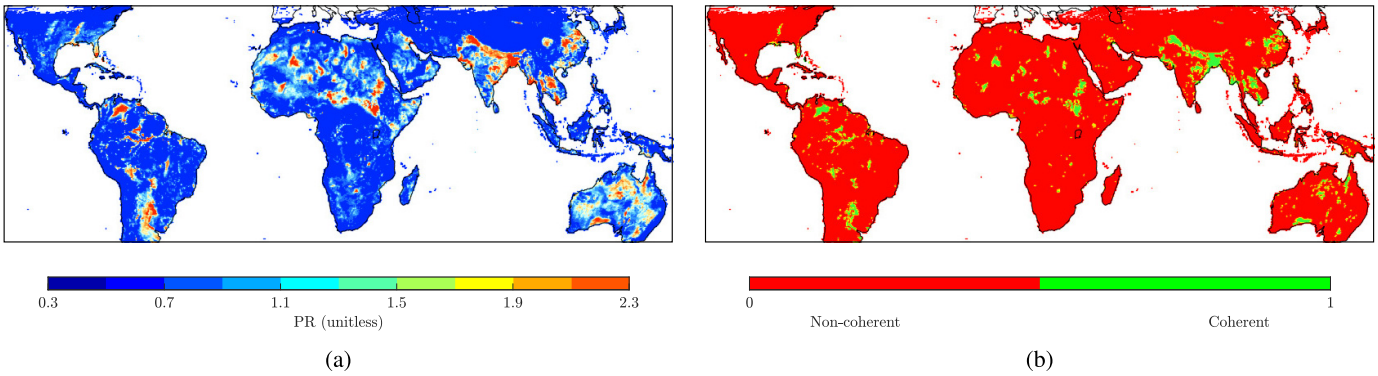


Fig. 7. Illustration highlighting persistent coherence over land points. (a) Median PR over sandbox-run 12-day period. (b) Median detection using $\rho_0 = 2.0$ over sandbox-run 12-day period.

showed terrain rms heights for such locations to be larger than those at locations regularly producing coherency.

C. Correspondence of DDM PR to SNR

The DDM SNR or the SNR corrected for antenna gain and range effects has been previously considered in land studies [18]–[20], with values that may be indicative of the nature of the observed return. Fig. 8 provides a scatterplot of PR versus the DDM SNR for the sandbox data set. For DDMs having $PR > 2.0$, the SNR is observed to range between 2.0 and approximately 40 dB, so that the DPSD provides information beyond that contained in the SNR alone. Cases having an SNR less than 1.5 dB were largely too noisy to allow clear classification; the high-exclusion thresholds used at low SNR (see Fig. 2) prevent detections from occurring in this case.

Approximately 2% of the measurements have $PR < 2.0$ but SNR values greater than 15 dB. Detailed examinations of such cases showed the DDMs typically to contain both coherent and incoherent contributions, and therefore not to fall into the "purely coherent" category. Future efforts could potentially consider a combination of the SNR and PR metrics to identify these "mixed" cases.

D. Probability of False Alarm

Evaluating the DPSD detector's probability of false alarm P_{fa} is difficult given the absence of truth information on the presence of coherence in the sandbox data set. However, the expectation that ocean surface DDMs should largely be incoherent [51]–[53] provides one opportunity for estimating P_{fa}. To test this, the DPSD was applied across a subset of ocean returns over the 12-day sandbox run period. The resulting P_{fa} was found to ranges $1.0 \times 10^{-5} \leq P_{\text{fa}} \leq 1.5 \times 10^{-5}$ for a threshold range of $1.8 \leq \rho_0 \leq 2.2$.

The subset used excluded areas with harbored waters, as discussed in Section IV-E. For the selected $\rho_0 = 2$ threshold, P_{fa} for the ocean subset is approximately 1.2×10^{-5}, and the typical range of PR is from 0.15 to 0.45. It is noted that this value for P_{fa} is small, so that the DPSD detector is unlikely to cause false alarms for "purely incoherent" scenes as typically occur for sea observations.

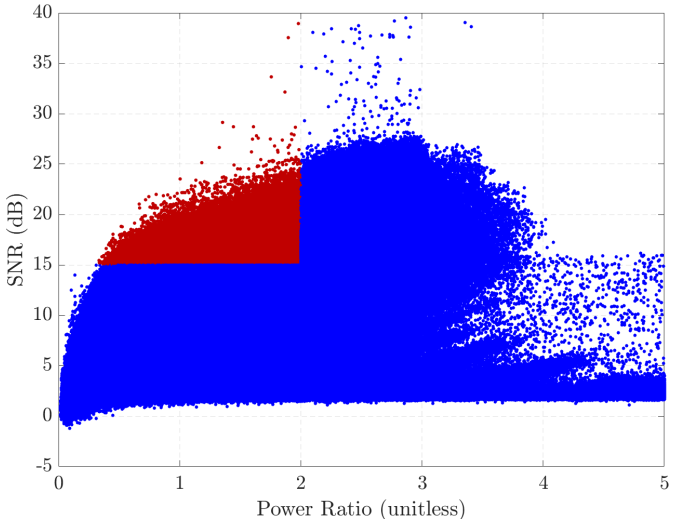


Fig. 8. Correspondence of SNR to DPSD PR for 16.5 million land specular points. Scatters in red are high-SNR low PR subset.

E. Ocean Coherency Detection

Although ocean measurements are expected to be incoherent, coherency can occur for sea returns when wind speeds are less than 3 m/s [39], [54] in regions that are sheltered from nonlocal swell waves. Coherence detections in ocean observations primarily occurred in a small set of sheltered locations; these locations were excluded from the ocean data subset used in assessing P_{fa}. The utility of the proposed detector can, therefore, also be extended to ocean surfaces. Examples of coherent detections in The Gulf region and the Maritime Continent are shown in Fig. 9 along with the corresponding mean wind speed obtained from the CYGNSS L3 wind retrievals during the 12-day data set. The results confirm that low wind speeds in sheltered areas can increase the likelihood of coherence for ocean observations. The Gulf and Maritime Continent regions account for 60% of the ocean coherence observed over the 12-day test data set. Detections typically occurred for sea areas enclosed between two or more land masses. Other locations where coherence was observed include the Alboran Sea, Gulf of Mexico, Caribbean Sea, and the Gulf of California. The locations over which coherence

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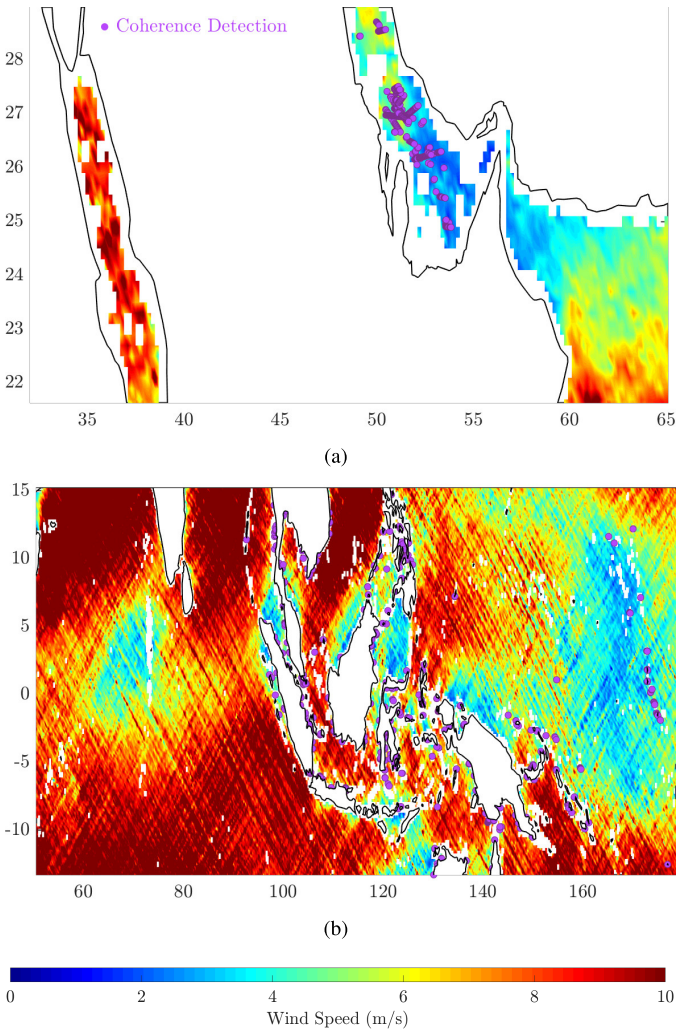


Fig. 9. Coherence detection over the ocean during a 12-day sandbox run overlaid on mean CYGNSS Level-3 winds. (a) Coherence detection over The Gulf region. (b) Coherence detection over Maritime Continent.

was observed over the ocean were, therefore, highly complementary to the regions with higher probabilities of calm conditions, that is, surface conditions conducive to coherence, as depicted in [55].

V. Conclusion

The DPSD algorithm proposed provides a method for identifying coherent returns in CYGNSS Level-1 DDM data for both land and ocean surfaces. Analysis of the 12-day data set suggests that approximately 8.9% of land returns are coherent. The results show that coherence is primarily associated with inland water bodies; approximately 80.7% of all coherent returns were associated with a nonzero water fraction within the associated Fresnel zone. Coherency was also observed in a limited set of arid desert regions for which it appears that surface roughness is sufficiently small. It was also shown that the proposed methodology can be used to identify coherent reflections in ocean observations.

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